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Characterization of eccentric hamstring strength and inter-limb asymmetry profiles in national-level youth male soccer players

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Abstract

Background Eccentric hamstring strength and inter-limb asymmetry are commonly monitored in youth football; however, evidence comparing their magnitude and directional consistency across age categories remains limited. This study compared eccentric hamstring strength and inter-limb asymmetry magnitude between national-level U12 and U18 male footballers and assessed within-cohort agreement in stronger-limb classification.

Methods In this cross-sectional study, 63 national-level male youth footballers were included: 35 U12 players aged approximately 12.0–12.9 years and 28 U18 players aged approximately 17.3–18.9 years. Participants completed repeated maximal bilateral Nordic Hamstring Exercise trials. Absolute peak force (PF), body mass-normalized peak force (RPF), inter-limb asymmetry magnitude, and stronger-limb classification were assessed. Trial-to-trial reliability was evaluated using coefficients of variation and intraclass correlation coefficients. Between-cohort differences in PF and RPF were examined using independent-samples t-tests, whereas the Mann–Whitney U test was used for inter-limb asymmetry. Hedges' g quantified between-cohort effect sizes, and Cohen's kappa assessed agreement in stronger-limb classification across trials.

Results The U18 cohort demonstrated greater absolute PF (383.1 ± 37.8 vs. 230.7 ± 46.6 N; g = 3.51, $p < 0.001$) and RPF (5.1 ± 0.5 vs. 4.5 ± 0.6 $N \cdot kg^{-1}$; g = 1.01, $p < 0.001$) than the U12 cohort. Inter-limb asymmetry was greater in U12 players (4.5 ± 2.3%) than in U18 players (3.3 ± 2.8%; g = −0.48, $p = 0.036$). The U12 cohort demonstrated substantial within-cohort agreement in stronger-limb classification ($\kappa = 0.860$, $p < 0.001$), whereas no evidence of agreement was observed in the U18 cohort ($\kappa = -0.143$, $p = 0.450$).

Conclusions National-level U18 male footballers exhibited greater absolute and body mass-normalized eccentric hamstring strength and lower mean inter-limb asymmetry than U12 players. The U12 cohort showed substantial within-cohort agreement in stronger-limb classification, whereas no evidence of agreement was observed in the U18 cohort. Strength magnitude, asymmetry magnitude, and asymmetry direction should therefore be interpreted

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as complementary monitoring indicators, preferably alongside repeated assessments rather than as interchangeable measures.

Keywords Nordic hamstring exercise, Peak force, Stronger-limb consistency, Trial-to-trial reliability, Body mass normalization, Age-group comparison

Introduction

Football match play is characterized by repeated high-intensity running and explosive actions, and these demands have increased over time in elite competition [1, 2]. In the English Premier League, high-intensity running distance increased by approximately 30% (890 ± 299 vs. 1,151 ± 337 m), while the number of high-intensity actions increased by approximately 50% (118 ± 36 vs. 176 ± 46) between the 2006–07 and 2012–13 seasons, despite an approximately 2% reduction in total distance covered [2]. Elite match play also requires repeated explosive sprints, rapid decelerations, and changes of direction [1, 2]. Eccentric hamstring strength is therefore commonly assessed in football and has been examined in relation to hamstring injury risk [3–6].

Despite widespread monitoring and injury-prevention strategies, hamstring strain injuries (HSIs) remain a major injury burden in men's professional football [7, 8]. In a 21-season prospective UEFA injury-surveillance study, 2,636 hamstring injuries accounted for 19% of all reported injuries, and the proportion increased from 12% in the 2001/02 season to 24% in the 2021/22 season [8]. Earlier epidemiological evidence identified the hamstrings as the most frequently injured muscle group, accounting for 37% of muscle injuries in professional football [7]. Although these epidemiological data derive mainly from adult professional football, age-specific assessment is also relevant in youth players because growth and biological maturation can influence strength, coordination, and injury-related characteristics. Players within the same chronological competition category may also be at different stages of biological maturation [9–11].

The Nordic Hamstring Exercise (NHE) is widely used to assess eccentric hamstring strength in football and is also included in strength-development and injury-prevention programmes [3, 12–15]. However, age-specific evidence in athletic youth remains limited [15–17]. Available studies suggest that knee-flexor strength and inter-limb asymmetry may differ across youth age categories, although findings depend on the population, testing mode, and method used to express asymmetry [15–19]. Repeated unilateral football actions, including kicking, accelerating, decelerating, and bracing, may contribute to limb-specific adaptations [18]. Asymmetry magnitude and direction are related but distinct characteristics: the size of the between-limb difference may remain small even when the stronger limb changes across trials or tasks [3, 20–22]. Evidence directly comparing eccentric strength, asymmetry magnitude, and stronger-limb consistency between younger and older elite male youth football cohorts remains limited [3, 15–17, 19, 21, 22].

This study aimed to compare eccentric hamstring strength, inter-limb asymmetry magnitude, and directional consistency of inter-limb asymmetry between national-level U12 and U18 male football cohorts using the NHE. We hypothesized that: (i) the U18 cohort would demonstrate greater absolute and body mass-normalized peak force than the U12 cohort; (ii) inter-limb asymmetry magnitude would be greater in the U12 cohort; and (iii) stronger-limb classification would show substantial agreement across repeated NHE trials in the U18 cohort but not in the U12 cohort.

Methods

Experimental approach to the problem

A cross-sectional observational comparison was conducted during the competitive season, consistent with previous age-category and elite-youth strength-asymmetry research [16, 17]. Testing was performed before weekday training sessions. Players were instructed to avoid high-intensity resistance exercise and match play for 48 h before testing and completed two supervised familiarization sessions during the preceding week [23].

Recruitment was not based on an a priori sample-size calculation. All eligible and available players from the two participating cohorts were included during the predefined testing period.

Subjects

Sixty-three national-level Chinese youth male footballers were included: a U12 cohort ($n = 35$; approximate age range: 12.0–12.9 years; all born in 2013), recruited from the 2025 Shanghai Elite Player Development Camps and regional competitions, and a U18 cohort ($n = 28$; approximate age range: 17.3–18.9 years; born between 2007 and 2008), comprising members of the Chinese National U18 Team. Participant age ranges were estimated from the available birth records and the date of testing. All participants competed at the highest domestic tier for their registered competition category and completed at least five football-specific sessions and one dedicated strength-and-conditioning session per week. Because the research team was providing scientific support to the national-team programme, only information required for the hamstring assessment was collected; detailed individual

football- and resistance-training histories and biological-maturation indicators were not recorded.

The experimental protocol received approval from the Ethics Committee of Shanghai University of Sport (No. 102772025RT222) and was conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from adult participants and from the parents or legal guardians of all minor participants before enrollment.

Nordic hamstring exercise test

Prior to assessment, participants performed a standardized warm-up protocol lasting 10–15 min [24]. This regimen commenced with 5 min of low-intensity cardiovascular priming (jogging) [25], followed by two sets of 10 repetitions for each dynamic lower-limb stretch, including walking lunges, lateral lunges, hip circles, and high knees [26]. To ensure specific neuromuscular activation, participants then executed three sub-maximal NHE practice trials at progressively increasing intensities (50%, 70%, and 90% of perceived maximum effort) [27]. These sub-maximal attempts were implemented by instructing participants to control their forward lean to a self-selected shallow, moderate, and deep range of motion (ROM) respectively, while using their hands to actively support and minimize the eccentric load during the final phase of each repetition [12].

Eccentric peak force was quantified using the NordBord hamstring testing system (NordBord V.1.0, VALD Performance, Albion, Australia), a field-based device commonly used to record bilateral NHE force [3, 15, 28]. The ankles were secured independently to the left and right force sensors, and force from both limbs was recorded simultaneously throughout each trial.

Participants were instructed to lean forward slowly and continuously at a self-selected, controlled pace while maintaining a straight alignment from the shoulders to the knees. With the arms crossed over the chest, they resisted the descent for as long as possible until they could no longer control the forward fall or reached the terminal position. Some participants completed the full range of motion without a clearly identifiable break point. No external load was applied, and the same standardized verbal instructions were used throughout. An assessor monitored trunk–thigh alignment, hip position, arm position, and descent control during every trial in accordance with established NHE testing procedures [3, 6, 15, 28, 29].

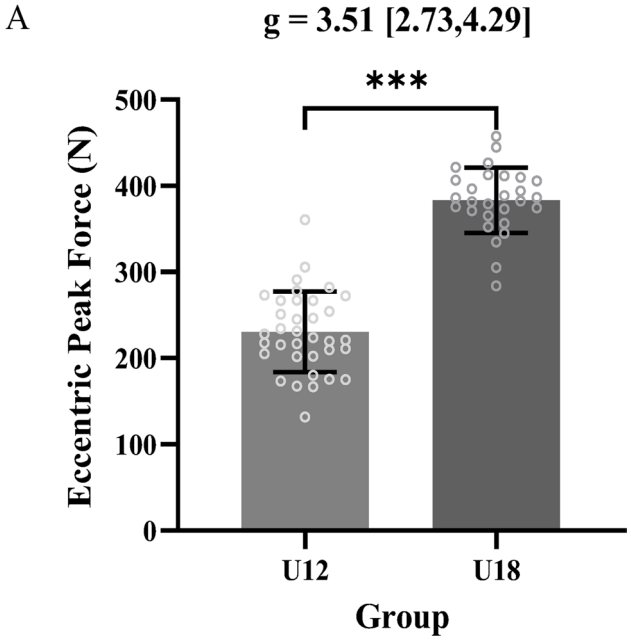
Each athlete initially completed two maximal trials within the same testing session, designated Trial 1 and Trial 2, with a 60-s rest interval. Repeated maximal NHE trials separated by short recovery periods are commonly used when measuring peak force [3, 13, 23, 28, 29]. No predetermined numerical threshold was used. A third trial was therefore administered when the assessor judged the first two force values to be inconsistent or when either trial deviated from the standardized technique, including visible hip flexion, loss of trunk–thigh alignment, premature movement of the arms, or failure to maintain a controlled forward descent; repeating technically invalid or unstable trials is consistent with published NHE testing practice [3, 13, 23]. When either Trial 1 or Trial 2 was deemed technically invalid, the invalid trial was excluded and replaced by the subsequent technically valid trial for reliability analysis. A numerical difference alone did not invalidate a trial; when both initial trials met the technical criteria, both remained in the reliability analysis. The primary outcomes were absolute PF (N) and body mass-normalized peak force (RPF; $N \cdot kg^{-1}$). RPF was included to facilitate comparison with youth NHE reference data and to reduce the influence of differences in body mass [15, 28].

Statistical analysis

Reliability analyses were based on the first two technically valid trials: Trial 1 and Trial 2 when both were valid, or the subsequent technically valid trial in place of an invalid trial. For between-cohort analyses, each participant's stronger- and weaker-limb PF values were taken from the technically valid trial containing the highest recorded PF value. Descriptive statistics are expressed as mean ± standard deviation (SD). Data normality was assessed using the Shapiro–Wilk test. Within-session trial-to-trial reliability between the two technically valid trials retained for analysis was assessed using the coefficient of variation (CV; [SD/mean] × 100%) and intraclass correlation coefficient (ICC; two-way mixed-effects, absolute agreement model). Reliability was considered acceptable when CV < 10% [30]. ICC values were classified as excellent (> 0.90), good (0.75–0.90), moderate (0.50–0.74), or poor (< 0.50) [31].

Between-group differences in PF and RPF were analysed using independent-samples t-tests, whereas the Mann–Whitney U test was used for non-normally distributed inter-limb asymmetry. Hedges' g effect sizes with 95% confidence intervals (CIs) were interpreted as trivial (0.00–0.19), small (0.20–0.59), moderate (0.60–1.19), large (1.20–1.99), or very large (> 2.00). A G*Power sensitivity analysis (two-sided $\alpha = 0.05$; 80% power; $n = 35$ and 28) indicated detectable standardized effects of $d = 0.722$ for the t-test and approximately $d = 0.739$ for the Wilcoxon–Mann–Whitney test; smaller effects were interpreted cautiously.

Because the NHE was performed bilaterally and force from both limbs was recorded simultaneously, inter-limb asymmetry was calculated relative to the summed force produced by both limbs, in accordance with Bishop et al. [21]:



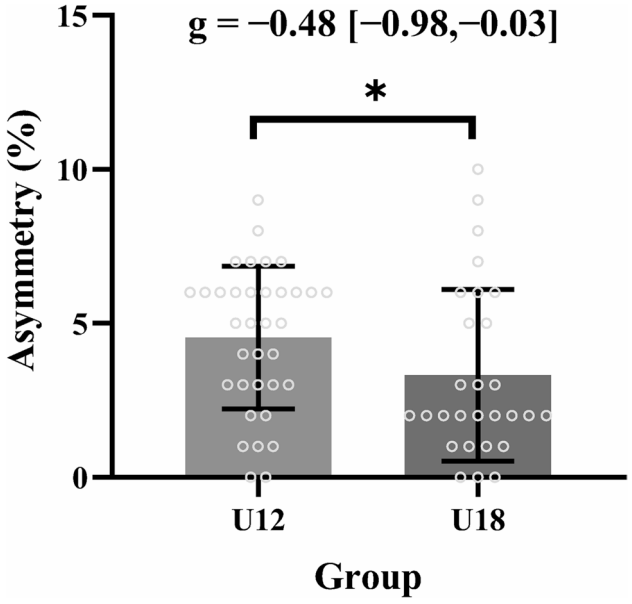


Fig. 2 Inter-limb asymmetry in eccentric hamstring strength. The y-axis denotes inter-limb asymmetry magnitude (%). Statistical parameters are displayed within the panel

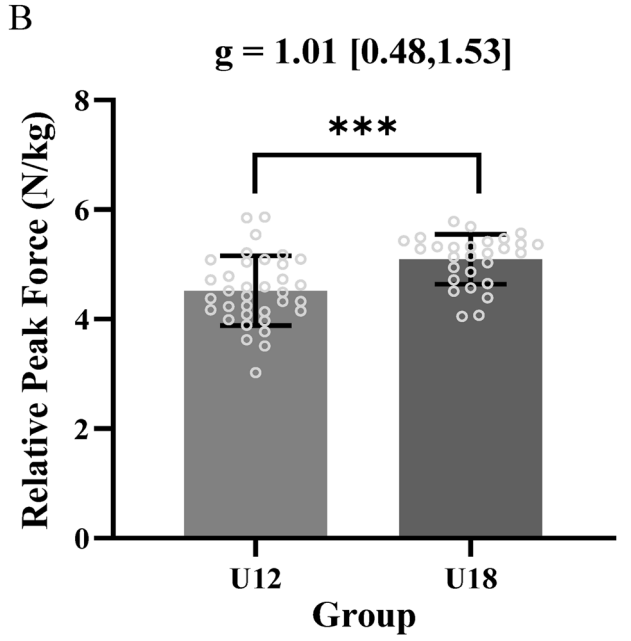


Fig. 1 Eccentric hamstring strength in U12 and U18 players. **A** Absolute eccentric peak force (PF; N) and (**B**) body mass-normalized peak force (RPF; $N{\cdot}kg^{-1}$). *** denotes a significant between-group difference at $p < 0.001$

$$\text{Asymmetry}\,(\%) = \left(\frac{\text{Stronger} - \text{Weaker}}{\text{Stronger} + \text{Weaker}}\right) \times 100\%$$

Asymmetry magnitude was defined as the percentage difference between the stronger and weaker limbs in NHE peak force, whereas directional consistency indicated whether the same limb was classified as stronger across repeated NHE trials. Inter-limb asymmetry was analyzed in terms of both magnitude and directional consistency. For each trial, the stronger limb was defined as the limb producing the higher eccentric peak force, independent of the participant's preferred kicking limb. Directional consistency was defined as whether the same limb was classified as stronger across the two technically valid trials retained for analysis. Cohen's kappa coefficient quantified agreement in stronger-limb classification between the retained trials [21, 22]. Statistical analyses were performed using SPSS version 27.0 (IBM Corp., Armonk, NY, USA), and data visualization was conducted in GraphPad Prism version 10 (GraphPad Software, San Diego, CA, USA). Statistical significance was set at $p < 0.05$.

Results

Levene's test confirmed homogeneity of variance for absolute eccentric peak force (PF; $F = 1.509$, $p = 0.224$) and body mass-normalized peak force (RPF; $F = 2.614$, $p = 0.111$). Independent-samples t-tests showed that the U18 cohort had greater absolute PF and RPF than the U12 cohort ($p < 0.001$; Fig. 1). Absolute PF was 383.1 ± 37.8 N in the U18 cohort and 230.7 ± 46.6 N in the U12 cohort. RPF was 5.1 ± 0.5 $N{\cdot}kg^{-1}$ in the U18 cohort and 4.5 ± 0.6 $N{\cdot}kg^{-1}$ in the U12 cohort.

Inter-limb asymmetry magnitude was $4.5 \pm 2.3\%$ in the U12 cohort and $3.3 \pm 2.8\%$ in the U18 cohort. The between-group difference was statistically significant ($p = 0.036$), with Hedges' $g = -0.48$ (Fig. 2).

Within-session trial-to-trial reliability metrics for bilateral eccentric peak force during the NHE are presented in Table 1. Cohen's kappa was used to assess agreement in stronger-limb classification between the two technically valid trials retained for analysis. The U12 cohort demonstrated substantial within-cohort agreement in stronger-limb classification across trials ($\kappa = 0.860$, $p < 0.001$). No evidence of within-cohort agreement was observed in the U18 cohort ($\kappa = -0.143$, $p = 0.450$), and stronger-limb classification varied between trials for some players (Fig. 3).

Discussion

The present study identified differences in eccentric hamstring strength and inter-limb asymmetry between national-level U12 and U18 male footballers. As hypothesized, the U18 cohort produced greater absolute and body mass-normalized peak force, whereas the U12 cohort showed greater inter-limb asymmetry. Contrary to the third hypothesis, the U12 cohort demonstrated substantial within-cohort agreement in stronger-limb classification, whereas no evidence of agreement was observed in the U18 cohort. Together, these findings suggest that strength level, asymmetry magnitude, and asymmetry direction do not necessarily vary in parallel. A small between-limb difference can therefore coexist with a change in the limb producing the higher value across repeated efforts [3, 21, 22].

The greater strength observed in the U18 cohort is consistent with reference data showing that absolute NHE strength increases across youth age categories, whereas body mass-normalized values differ much less from U14 onward [15]. The present RPF values of 4.5 $N \cdot kg^{-1}$ in U12 and 5.1 $N \cdot kg^{-1}$ in U18 players were within or above the approximately 3.6–4.7 $N \cdot kg^{-1}$ range reported across U13–U18 groups by Jeanguyot et al. [15]. The between-cohort difference remained after normalization, suggesting that body mass alone may not account for the observed strength gap. Biological maturation, increases in lean tissue, and accumulated exposure to football and resistance training may also contribute, because strength and power gains during adolescence reflect both body-size changes and maturation- and training-related neuromuscular adaptation [9, 32, 33]. Body mass is associated with NHE force; therefore, reporting PF and RPF together provides complementary information about absolute and body-mass-normalized strength [15, 28].

The greater mean asymmetry in the U12 cohort (4.5% vs. 3.3%) provides additional context for the strength findings. Kalata et al. [16] similarly reported a higher incidence of knee-flexor asymmetry in younger age categories followed by a reduction in later adolescence, although their isokinetic protocol differs from the NHE used here. Repeated unilateral actions in football may create athlete-specific loading histories, while growth, coordination, and training experience may influence how force is distributed between limbs [9, 16, 18]. Thus, the lower average asymmetry observed in the U18 cohort may reflect a more balanced expression of eccentric force within this high-performance group. Nevertheless, the absolute group means were small and should not be converted into a universal injury-risk threshold. These values may be most useful as age-category-specific references against which persistent individual profiles can be monitored over time [3, 21].

Directional consistency can provide information that is not captured by asymmetry magnitude alone [3, 21–23]. The U12 cohort showed substantial agreement in the stronger limb across the two technically valid trials retained for analysis ($\kappa = 0.860$), whereas no evidence of agreement was observed in U18 players ($\kappa = -0.143$, $p = 0.450$). Similar variability has been reported in professional footballers tested with the NordBord, where agreement in the stronger limb ranged from 43% to 57% across sessions despite generally acceptable CVs [3]. When between-limb differences are small, modest biological or technical variation can reverse which limb is classified as stronger without materially changing overall asymmetry magnitude [21, 22]. The present reliability results are consistent with this interpretation: U18 CVs were acceptable (5.17–6.10%), while ICCs were poor to moderate (0.432–0.686). In a relatively homogeneous national-level squad, restricted between-player variance can reduce ICC even when absolute trial-to-trial variation is modest

Table 1 Within-session trial-to-trial reliability of bilateral eccentric hamstring peak force during the Nordic Hamstring Exercise

Variables	U12		U18	
	CV	ICC	CV	ICC
PF (L)	4.21% (3.23% – 5.20%)	0.936 (0.877–0.967)	5.17% (3.89% – 6.45%)	0.686 (0.426–0.841)
PF (R)	5.54% (3.53% – 7.55%)	0.887 (0.788–0.941)	6.10% (4.10% – 8.09%)	0.524 (0.194–0.747)
RPF (L)	4.23% (3.24% – 5.21%)	0.878 (0.772–0.937)	5.20% (3.92% – 6.47%)	0.662 (0.389–0.828)
RPF (R)	5.53% (3.53% – 7.54%)	0.808 (0.652–0.898)	6.09% (4.09% – 8.09%)	0.432 (0.077–0.690)

PF Peak force; *RPF* Body mass-normalized peak force, *CV* Coefficient of variation, *ICC* Intraclass correlation coefficient (two-way mixed-effects, absolute agreement model)

All CV values remained < 10%, demonstrating that the absolute reliability of the measurements met the established threshold for acceptability. For the U12 cohort, all ICC values exceeded 0.75, indicating good-to-excellent relative reliability. Conversely, ICC values for the U18 cohort ranged from 0.432 to 0.686, reflecting poor-to-moderate relative reliability

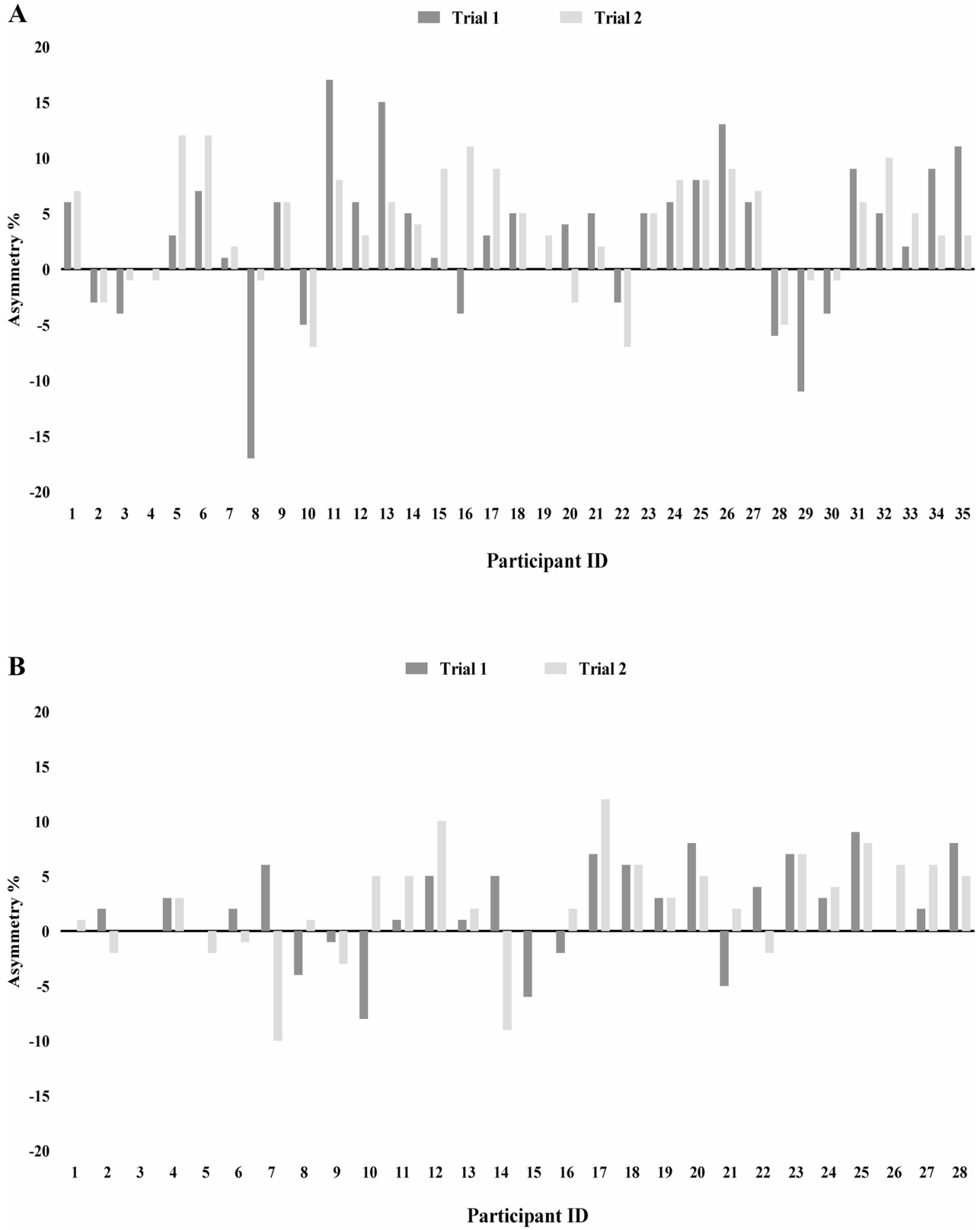


Fig. 3 Trial-to-trial consistency in the direction and magnitude of inter-limb asymmetry for individual participants. Panel **A** presents the U12 cohort ($n = 35$), and Panel **B** presents the U18 cohort ($n = 28$). The x-axis shows participant identification numbers (1–35 for U12 and 1–28 for U18). Trial 1 and Trial 2 denote the two technically valid NHE trials retained for analysis. When an initial attempt was technically invalid, it was excluded and replaced by the subsequent technically valid trial; retained trials were separated by at least 60 s. Positive values indicate greater peak force in the right limb, whereas negative values indicate greater peak force in the left limb; a change in sign indicates that the limb classified as stronger changed between trials

[31]. The stronger-limb direction may therefore vary over time, particularly when asymmetry magnitude is close to measurement error.

These findings should be interpreted in light of several limitations. The cross-sectional comparison cannot establish within-player developmental trajectories, and the absence of stature, biological-maturation status, playing position, and detailed individual training exposure prevented these factors from being modelled. Ratio scaling assumes an approximately proportional relationship between force and body mass. Allometric scaling was not performed, and the between-cohort RPF difference should therefore be interpreted cautiously. Descent velocity was not quantified kinematically. Finally, the standard protocol comprised two maximal trials, with a third trial used only when technique or consistency was questioned by the assessor rather than by a prespecified numerical criterion. These limitations increase uncertainty around the mechanisms underlying the observed cohort differences and should be considered when interpreting the findings. Longitudinal studies incorporating maturation, training exposure, and repeated strength assessments are needed to determine how individual strength and asymmetry profiles change through adolescence.

Practical implications

Practitioners should interpret NHE results using several measures rather than relying on a single asymmetry cut-off. Absolute PF describes eccentric force capacity, RPF expresses peak force normalized to body mass, and repeated trials show whether the same limb is consistently stronger. Because mean asymmetry was small in both cohorts, an isolated value should be interpreted alongside test reliability and the player's previous results.

Regular NHE testing can help determine whether a between-limb difference persists over time or reflects normal trial-to-trial variation. Training decisions should combine repeated NHE results with injury history, current training exposure, maturation information when available, and other performance assessments.

Conclusion

National-level U18 male footballers showed greater absolute and body mass-normalized eccentric hamstring strength than U12 players, whereas the U12 cohort had a higher mean inter-limb asymmetry. The U12 cohort demonstrated substantial within-cohort agreement in stronger-limb classification across repeated trials, whereas no evidence of agreement was observed in the U18 cohort. These findings suggest that higher strength and lower asymmetry magnitude do not necessarily coincide with a consistent directional preference. Considering PF, RPF, asymmetry magnitude, and directional consistency together may provide a broader profile for age-category-specific monitoring. Longitudinal research incorporating biological maturation and detailed training exposure would help determine how these characteristics evolve within individual players.

Abbreviations

PF	Peak force
RPF	Body mass-normalized peak force
NHE	Nordic Hamstring Exercise
HSI	Hamstring strain injury
COD	Changes of direction
PHV	Peak height velocity
ROM	Range of motion
CV	Coefficient of variation
ICC	Intraclass correlation coefficient

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Clinical trial number

Clinical trial number: not applicable.

Authors' contributions

Y. L, J. W, and D. J conceived and designed the research. J. W, C. Z, Z. B, and Y. T conducted the experiments. J. W analyzed the data. J. W, M. L, H. Z, D. J, C. B, and Y. L wrote the manuscript. All authors have read and approved the final manuscript.

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Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study was approved by the Ethics Committee of Shanghai University of Sport (Approval No. 102772025RT222). Written informed consent was obtained from adult participants and from the parents or legal guardians of all minor participants before enrollment.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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